

Lecture 2 — Image Representation

Learning outcomes

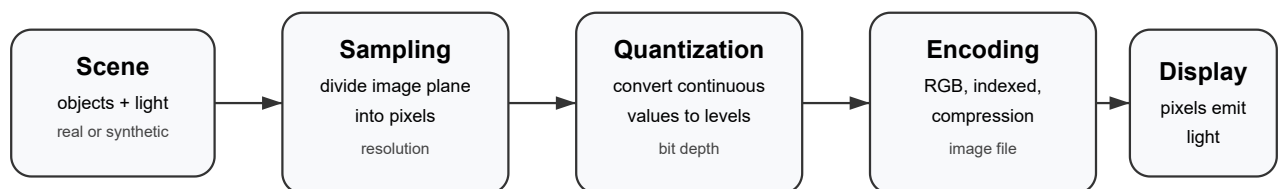
1. Explain the idea of a raster image as a two-dimensional array of pixels.
 2. Distinguish between sampling and quantization.
 3. Describe the RGB color model and calculate basic color-depth values.
 4. Explain direct color coding and indexed color using lookup tables.
 5. Identify common image file formats and their appropriate uses.
 6. Explain the basic roles of display monitors and printers in image reproduction.
 7. Relate image representation to later computer graphics topics such as scan conversion, anti-aliasing, rendering, texture mapping, and image processing.
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1. Why image representation matters

Computer graphics eventually becomes an image. Even if the original scene contains three-dimensional objects, lights, cameras, curves, surfaces, materials, and transformations, the final output on a normal display is usually a rectangular grid of pixels.

So before studying line drawing, clipping, transformation, projection, hidden surface removal, or animation, we need to understand what kind of object the computer is finally producing.

Image Representation: from light to numbers to display



A digital image is a structured table of numbers

Each pixel stores attributes such as color, opacity, depth, or labels.
Computer graphics algorithms decide what these values should be.
Image-processing algorithms modify or interpret the values after they exist.

A digital image is not only a picture. It is also a **data structure**.

At the lowest level, a simple image may be stored as a table such as:

```
image[y][x] = pixel value
```

Depending on the image type, that pixel value may represent:

- black or white,
- a grayscale intensity,
- an RGB color,
- an index into a color table,
- an opacity value,
- a depth value,
- a label or object identifier,
- or even several of these at once.

This is a powerful idea: **an image is a matrix of meaningful numbers.**

Classroom thought

A photograph of a fish, a screenshot of a mobile app, a satellite image of Bangladesh, and a game frame may look very different. But inside the computer, all of them can be represented as grids of numerical samples.

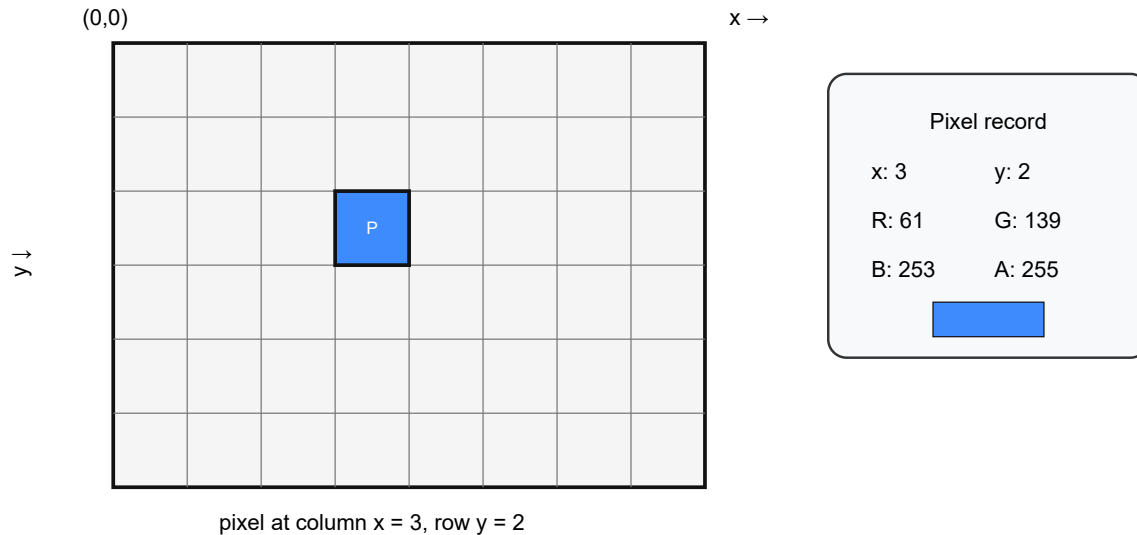
That common representation is why similar algorithms can be used in photography, graphics, medical imaging, remote sensing, computer vision, and digital art.

2. Raster image as a pixel grid

A **raster image** is a rectangular arrangement of small elements called **pixels**. The word pixel comes from *picture element*.

A pixel is the smallest addressable unit of a raster image. It has a position and one or more stored values.

A raster image as a 2D array of pixels



An image file stores millions of such records compactly.

If an image has width **W** and height **H**, then it contains:

Number of pixels = $W \times H$

For example:

$1920 \times 1080 = 2,073,600$ pixels

That is why a 1920×1080 image is commonly called a **2-megapixel** image.

Coordinate convention

In many image systems, the top-left pixel is **(0, 0)**.

- **x** increases from left to right.
- **y** increases from top to bottom.

This is different from the usual mathematical coordinate system, where the y-axis increases upward. This difference becomes important when drawing lines, placing objects, handling mouse input, and converting between mathematical and screen coordinates.

Important distinction

A mathematical point has no size. A pixel has area.

A mathematical line has infinitely many points. A raster line is a finite pattern of pixels.

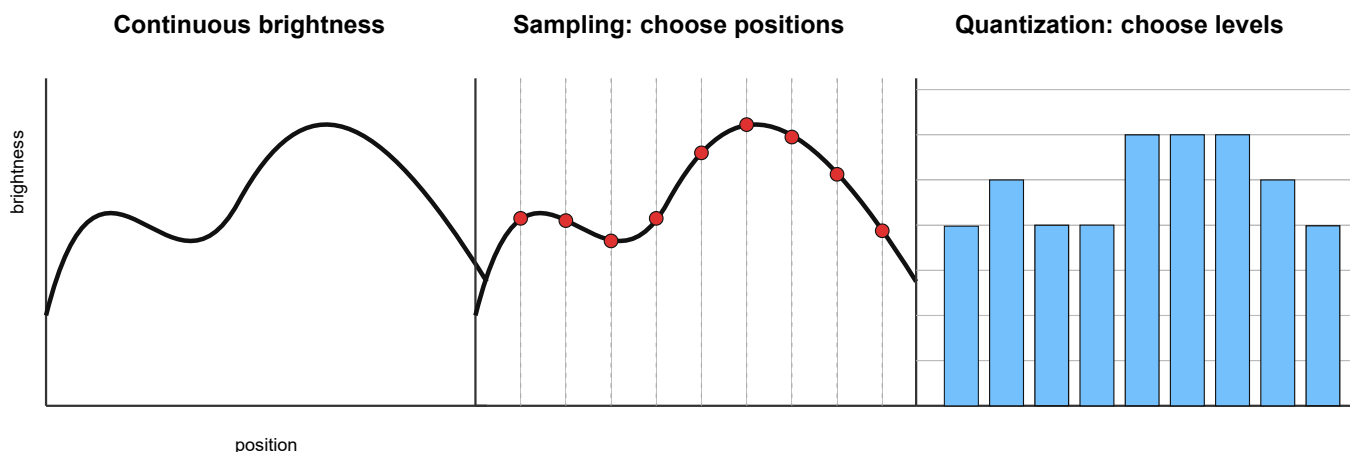
This gap between continuous mathematics and discrete pixels is one of the central issues of computer graphics.

3. Sampling and quantization

When a real or synthetic scene becomes a digital image, two important processes occur:

1. Sampling
2. Quantization

Sampling and quantization



Sampling controls spatial detail; quantization controls value precision. Too little of either creates visible artifacts.

3.1 Sampling

Sampling means measuring a continuous signal at discrete positions.

In an image, sampling decides **where** we measure the scene. For a raster image, the scene is sampled at pixel positions.

Higher sampling density means more pixels per unit area. This usually allows more spatial detail.

For example:

```
640 × 480      = 307,200 pixels
1920 × 1080    = 2,073,600 pixels
3840 × 2160    = 8,294,400 pixels
```

A 4K UHD image has four times as many pixels as a Full HD image because both width and height are doubled.

3.2 Quantization

Quantization means converting a continuous value into one of a finite number of levels.

For example, suppose brightness can theoretically vary continuously from black to white. A digital grayscale image may store brightness using only 8 bits:

```
0    = black
255 = white
```

So there are only 256 possible brightness levels.

3.3 Sampling error and quantization error

If sampling is too low, fine details may disappear or produce aliasing artifacts.

If quantization is too coarse, smooth gradients may become visible steps. This is called **banding**.

Trivia: why old games looked blocky but charming

Many old computer and console games had low resolution and limited color palettes. This was not only an artistic choice; it was a hardware limitation. Memory was expensive, processors were slower, and displays were less capable. Modern pixel art deliberately imitates these limitations as a style.

4. Pixel attributes

A pixel may store more than just color.

Common pixel-related attributes include:

Attribute	Meaning	Example use
Intensity	Brightness value	grayscale images
RGB	red, green, blue channels	display images
Alpha	transparency or opacity	UI, compositing, PNG images
Depth / Z	distance from camera	3D rendering, hidden surface removal
Stencil	mask or special control value	rendering effects
Label	object or region identifier	segmentation, computer vision

In simple introductory graphics, we usually begin with RGB color. But in real rendering systems, each pixel may be associated with multiple buffers:

- color buffer,
- depth buffer,
- stencil buffer,
- accumulation buffer,
- normal buffer,
- object ID buffer.

These buffers help the graphics system decide what the final image should look like.

5. The RGB color model

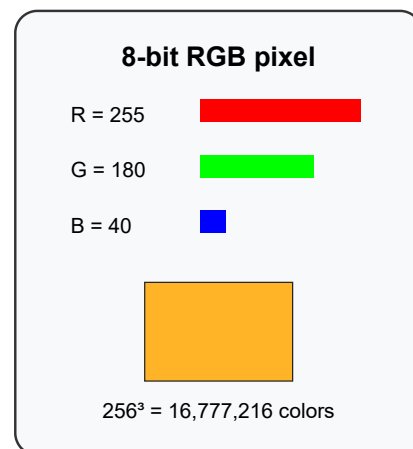
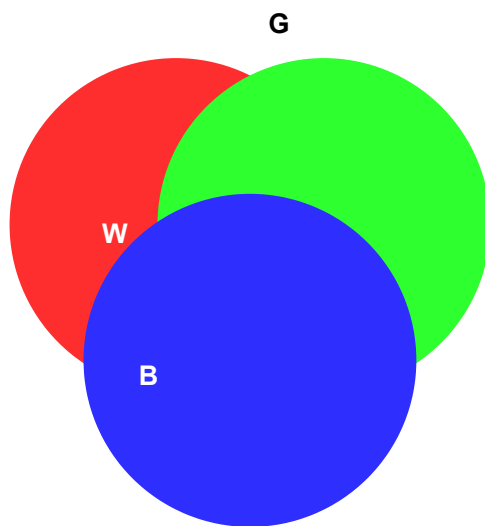
The most common color model for displays is **RGB**.

RGB means:

- Red,
- Green,
- Blue.

A display creates many colors by combining different amounts of red, green, and blue light.

RGB color model: additive mixing of light



On displays, adding red, green and blue light creates brighter colors; all three at maximum create white.

For an 8-bit-per-channel RGB image, each channel can store 256 possible values:

```
Red    ∈ [0, 255]
Green  ∈ [0, 255]
Blue   ∈ [0, 255]
```

Therefore, the total number of possible colors is:

```
256 × 256 × 256 = 16,777,216 colors
```

This is called **24-bit color**, because:

```
8 bits for red + 8 bits for green + 8 bits for blue = 24 bits
```

Some important RGB values

RGB value	Color
(0, 0, 0)	black
(255, 255, 255)	white
(255, 0, 0)	red
(0, 255, 0)	green
(0, 0, 255)	blue
(255, 255, 0)	yellow
(255, 0, 255)	magenta
(0, 255, 255)	cyan

Trivia: green gets special treatment in many systems

Human vision is very sensitive to green detail. Some image and video systems exploit this fact. For example, many camera sensors use a Bayer pattern with more green samples than red or blue samples. Many video compression methods also preserve brightness information more carefully than color information because human perception notices brightness detail strongly.

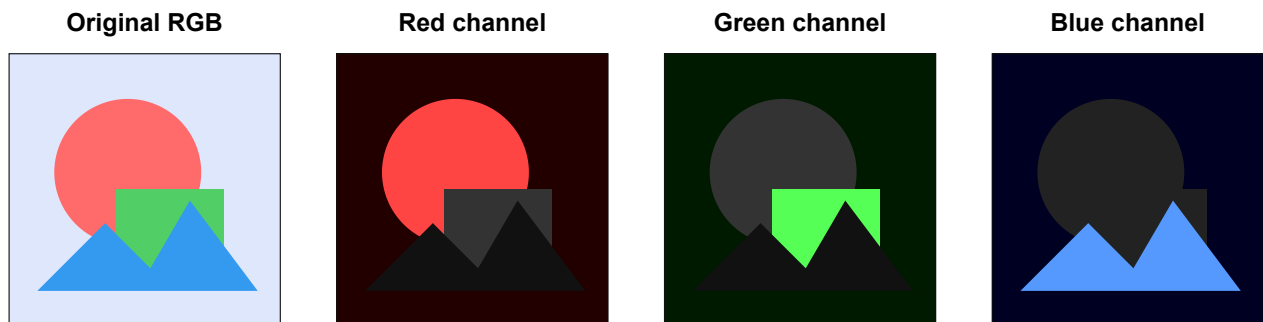
6. Color channels

An RGB image can be thought of as three grayscale images stacked together:

1. Red channel,

2. Green channel,
3. Blue channel.

Separating an image into color channels



An RGB image can be viewed as three grayscale images stacked together: one for red, one for green, and one for blue.

Many graphics and image-processing algorithms work on channels separately.

For example, if a pixel has:

```
R = 200  
G = 50  
B = 30
```

then the pixel is mostly red, with smaller amounts of green and blue.

Separating channels is useful in many operations:

- changing color balance,
- extracting features,
- applying filters,
- compressing images,
- processing satellite images,
- detecting objects in computer vision.

Application: medical and scientific images

Not every image channel has to mean red, green, and blue. In scientific visualization, channels may represent different measurements. For example, a false-color image may use red to show

high temperature, blue to show low temperature, and green to show another measured property.

In that case, color is not just decorative. It becomes a language for communicating data.

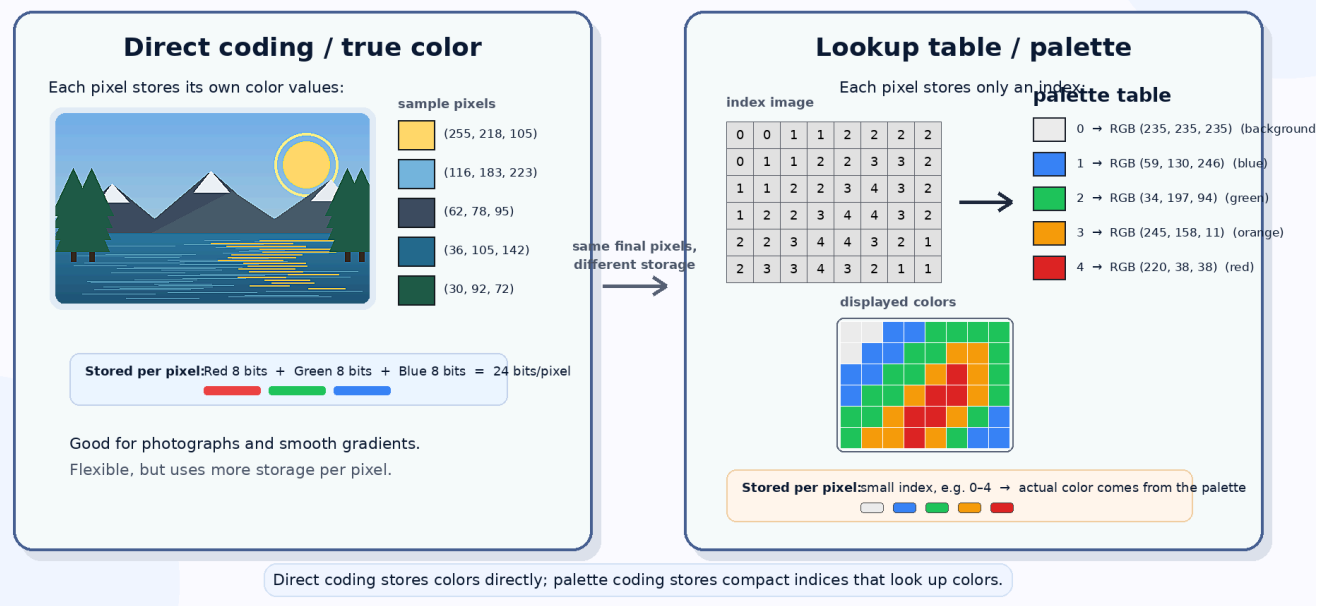
7. Direct coding and lookup tables

There are two common ways to store pixel color:

1. Direct color coding
2. Indexed color using a lookup table

Direct color coding vs lookup table

Two ways to store the color of pixels in a raster image.



7.1 Direct color coding

In direct coding, each pixel stores the actual color values.

Example:

Pixel = (R, G, B) = (120, 200, 80)

This is flexible because each pixel can choose any available color.

For a 24-bit RGB image:

bits per pixel = 24

```
bytes per pixel = 3
```

So an uncompressed 1920×1080 RGB image requires approximately:

```
1920 × 1080 × 3 = 6,220,800 bytes  
≈ 5.93 MiB
```

That is only one frame. A video contains many frames per second, so compression becomes essential.

7.2 Lookup table / palette-based color

In indexed color, each pixel stores a small integer index. That index points to a color in a table called a **palette** or **lookup table**.

Example:

```
Pixel value = 3  
Palette[3] = (255, 128, 0)
```

This means the pixel is orange.

If an image uses only 256 colors, each pixel needs only 8 bits for the index. The actual RGB colors are stored separately in the palette.

Why use a lookup table?

A lookup table can save memory when the number of colors is small. It can also make color changes very fast. If many pixels refer to the same palette entry, changing that entry changes all of those pixels visually.

Trivia: palette animation

Some older games created animation effects by changing only the color palette, not the pixel indices. Waterfalls, glowing lights, fire, and magic effects could be animated very cheaply by cycling colors in the palette.

This is a beautiful example of using representation cleverly to reduce computation.

8. Bit depth

Bit depth describes how many bits are used to represent each pixel or each channel.

Bit depth and tone levels

1-bit grayscale: 2 levels



2-bit grayscale: 4 levels



4-bit grayscale: 16 levels



8-bit grayscale: 256 levels

Higher bit depth reduces banding and allows smoother gradients.



For grayscale images:

Bits per pixel	Number of levels
1 bit	2 levels
2 bits	4 levels
4 bits	16 levels
8 bits	256 levels
10 bits	1024 levels
12 bits	4096 levels
16 bits	65536 levels

For RGB images, bit depth may be described per channel or per pixel.

Examples:

```
8-bit grayscale      = 8 bits per pixel
8-bit RGB per channel = 24 bits per pixel
8-bit RGBA per channel = 32 bits per pixel
```

RGBA adds an alpha channel:

```
R + G + B + A
```

where **A** usually represents opacity.

Why higher bit depth matters

Higher bit depth is useful in:

- medical imaging,
- HDR photography,
- cinema production,
- scientific visualization,
- image editing,
- physically based rendering.

If an image will be heavily edited, higher bit depth gives more room for adjustment before artifacts appear.

Quick calculation

How much memory is needed for an uncompressed 4K UHD 8-bit RGB image?

```
Width   = 3840
Height  = 2160
Bytes per pixel = 3

Memory = 3840 × 2160 × 3
        = 24,883,200 bytes
        ≈ 23.7 MiB
```

At 60 frames per second, uncompressed data rate would be approximately:

```
23.7 MiB × 60 ≈ 1422 MiB/s
```

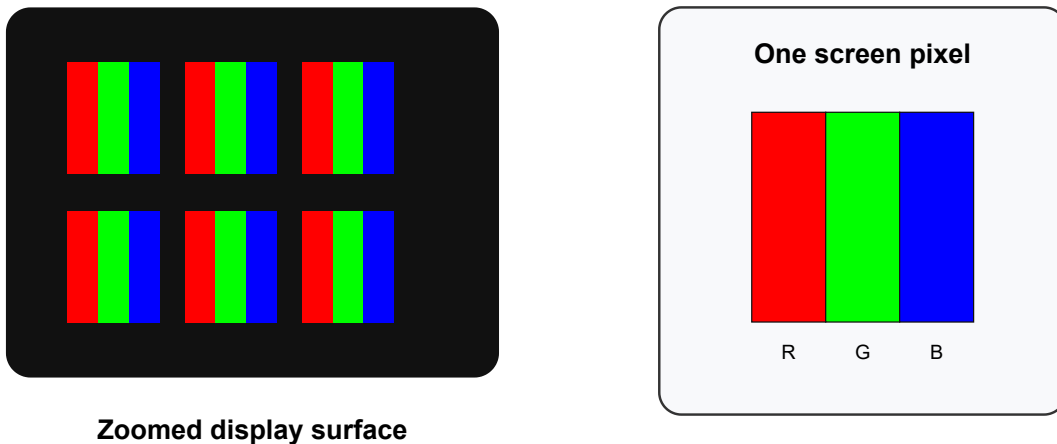
This is why modern graphics and video systems rely heavily on specialized hardware, memory bandwidth, and compression.

9. Display monitors

A monitor converts digital image data into visible light.

Although we usually talk about pixels abstractly, real monitors have physical subpixels.

Display pixels and RGB subpixels



The eye usually blends tiny red, green and blue emitters into one perceived color.

A typical color display pixel is composed of red, green, and blue subpixels. By controlling their intensities, the display creates the perception of a single color.

Display resolution

Display resolution is the number of addressable pixels.

Examples:

Name	Resolution
HD	1280 × 720
Full HD	1920 × 1080
QHD	2560 × 1440
4K UHD	3840 × 2160
8K UHD	7680 × 4320

Pixel density

Pixel density measures how many pixels are packed into a physical length, often expressed as PPI: pixels per inch.

A mobile phone screen may have a much higher pixel density than a classroom projector. Therefore, the same image may appear sharper on the phone even if the file has the same pixel

dimensions.

Display technologies

Different display technologies create light in different ways.

How common display technologies create images

Different hardware, same computer-graphics abstraction: pixels, color, and brightness.



Computer graphics abstracts hardware as pixels, but real displays strongly affect brightness, contrast, color, motion, and energy use.

Common display technologies include:

- **CRT:** electron beam scans a phosphor-coated screen.
- **LCD:** a backlight shines through liquid crystal filters.
- **OLED:** each pixel emits its own light.
- **MicroLED:** tiny LEDs act as individual light-emitting pixels.

Trivia: why black looks different on different screens

On many LCD displays, black is produced by blocking a backlight. Some light may still leak through, so black may look dark gray. On OLED displays, black pixels can be turned off, producing much deeper black. This is why the same image can look different on different screens.

10. Printers and hard-copy output

A printer converts digital image data into marks on paper.

Screens and printers are fundamentally different:

- A screen emits light.
- Paper reflects light.

That is why screen color and print color use different models.

RGB

for Digital Screens

Best for Digital Screens (Ph, Tab, Comp)



●
R

●
G

●
B

RGB for Digital Screens (Ph, Tab, Co
Best for digital screens like
phones, tablets, computers.

CMYK

for Printing

Printed on highquality matte paper)



●
C

●
M

●
Y

●
K

CMYK uses Cyan, Magenta,
Yellow, Black ink.
Best for printing on paper.

Note: RGB is bright and vibrant for screens,
CMYK is optimized for accurate printing.

Displays commonly use RGB. Printers commonly use **CMYK**:

- Cyan,
- Magenta,
- Yellow,
- Key/Black.

RGB is additive. CMYK is subtractive.

Why printed colors may not match the monitor

A color may look bright on screen because the screen emits light. The same color may look dull on paper because paper can only reflect available light and ink absorbs part of it.

Other factors also matter:

- printer type,
- paper type,
- ink quality,
- color profile,
- room lighting,
- monitor calibration.

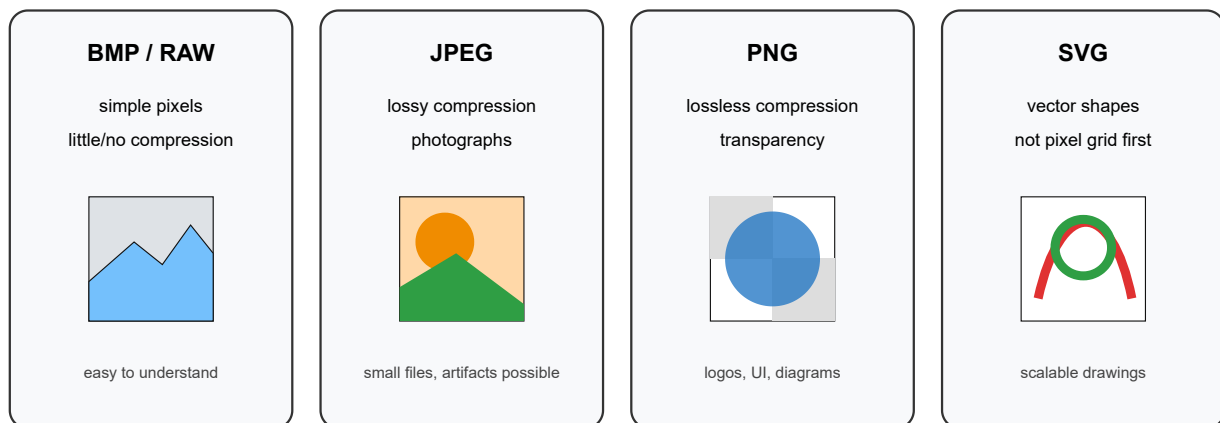
Application: graphic design and publishing

Designers often work in RGB for digital content and convert to CMYK for print production. This conversion can shift colors. Professional workflows use color management systems to reduce surprises.

11. Image files

An image file stores image data in a particular format. Different formats are designed for different needs.

Common image file formats: what they are good for



Choosing a format is an engineering decision: quality, size, transparency, editability and compatibility all matter.

11.1 BMP and raw formats

BMP and raw formats are conceptually simple. They store pixel data with little or no compression.

Advantages:

- easy to understand,
- useful for teaching,
- sometimes useful in low-level processing.

Disadvantages:

- large file size,
- not ideal for web distribution.

11.2 JPEG

JPEG is commonly used for photographs. It uses lossy compression, meaning some information is discarded to reduce file size.

Advantages:

- very small files for natural images,
- widely supported.

Disadvantages:

- compression artifacts,
- poor for sharp text, diagrams, and repeated editing.

11.3 PNG

PNG uses lossless compression and supports transparency.

Advantages:

- good for diagrams, icons, screenshots, and UI graphics,
- preserves sharp edges,
- supports alpha transparency.

Disadvantages:

- may be larger than JPEG for photographs.

11.4 GIF

GIF supports indexed color and simple animation.

Advantages:

- simple short animations,
- wide compatibility.

Disadvantages:

- limited color palette,
- not ideal for high-quality images.

11.5 SVG

SVG is a vector format. It stores shapes, paths, text, and styling rather than a fixed pixel grid.

Advantages:

- scalable,
- editable,
- excellent for diagrams, icons, and logos.

Disadvantages:

- not suitable as the main format for ordinary photographs.

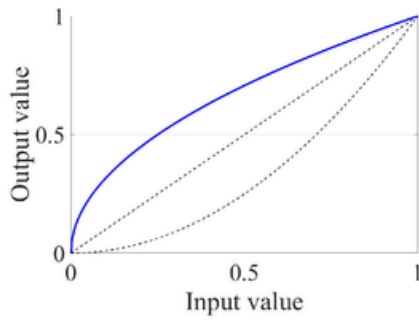
Practical rule

Use:

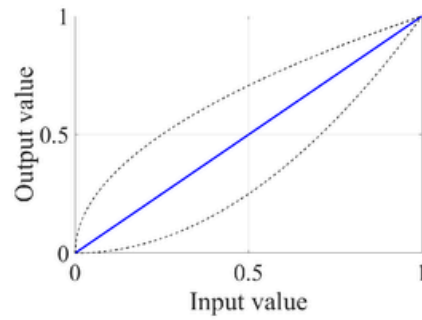
- JPEG for photographs,
- PNG for screenshots, diagrams, transparency, and UI assets,
- SVG for icons, logos, and diagrams,
- GIF or video formats for short animations,
- specialized formats for medical, scientific, or professional imaging.

12. Gamma and perception

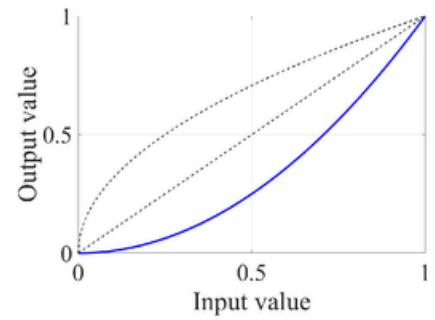
We often assume that pixel value 128 means exactly half as bright as pixel value 255. In real imaging systems, the situation is more complex because displays, cameras, image formats, and human vision involve nonlinear relationships.



(a)



(b)

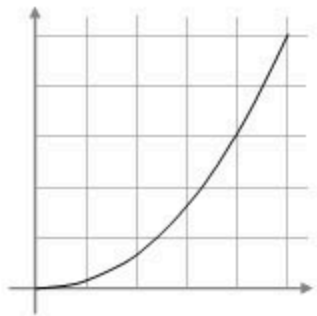


(c)

Gamma correction is used to manage the nonlinear relationship between stored pixel values and displayed brightness.

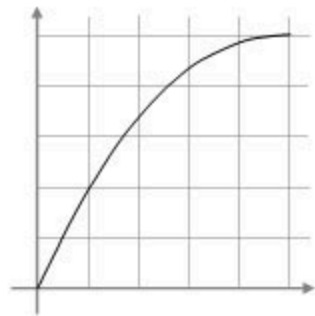
This matters in graphics because many operations should be performed carefully:

- color blending,
- shading,
- antialiasing,
- texture filtering,
- image compositing,
- physically based rendering.



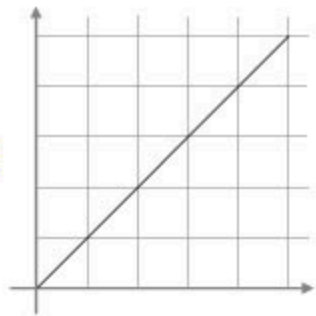
Gamma characteristics of monitors

×



Color information adjusted to match gamma characteristics

=



Color handling approaching the "y = x" ideal

Simple intuition

Human vision does not perceive brightness linearly. We are often more sensitive to relative changes in darker regions than equal physical changes in very bright regions. Imaging systems take advantage of this perceptual behavior.

Trivia: the middle gray surprise

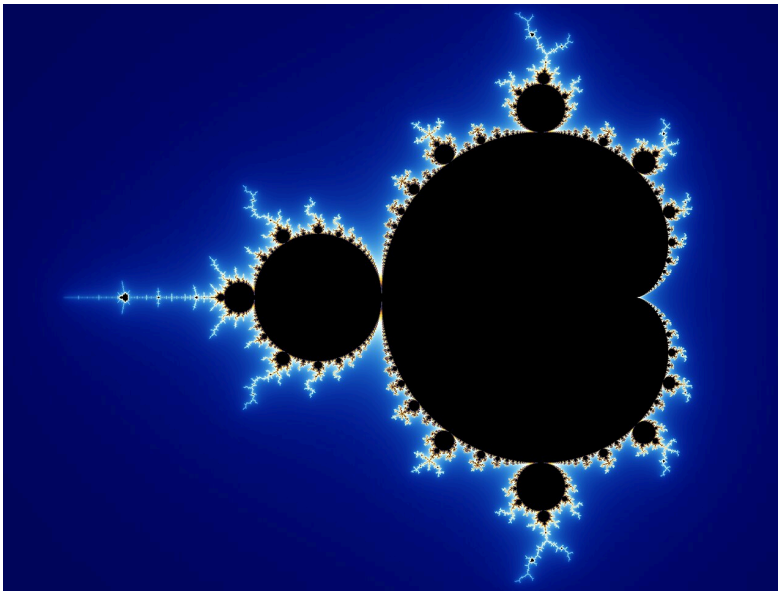
In many digital systems, a stored value around 128 does not correspond to half the physical light output of 255. This surprises beginners, but it is one reason image processing and rendering software needs to treat color spaces carefully.

13. Visualizing the Mandelbrot set

The course outline mentions visualizing the Mandelbrot set. This is an excellent example of image representation because the image is generated by computation rather than photography.

Visualizing the Mandelbrot set: pixels from repeated computation

A famous bridge between math and images



For each pixel

1. Map pixel to a complex number c
2. Start $z = 0$
3. Repeat $z \leftarrow z^2 + c$
4. Count escape speed
(how fast z diverges)
5. Convert count to a color

The image is not photographed.
It is computed pixel by pixel.

For each pixel:

1. Map the pixel position to a point in the complex plane.
2. Use that point as a complex number c .
3. Start with $z = 0$.
4. Repeatedly compute:

$$z = z^2 + c$$

5. If z escapes beyond a threshold, color the pixel based on how quickly it escaped.
6. If it does not escape within a chosen number of iterations, color it as part of the set.

[Video: coding mandelbrot set in Python](#)

Why this is fascinating

A very simple formula creates an extremely complex visual structure. Zooming into the Mandelbrot set reveals repeated patterns, spirals, miniature copies, and endless boundary detail.

This shows one of the joys of computer graphics: **small mathematical rules can produce rich visual worlds.**

Application connection

Fractals and procedural patterns are used in:

- terrain generation,
 - cloud and smoke textures,
 - natural-looking surfaces,
 - visual effects,
 - mathematical art,
 - educational visualization.
-

14. Image representation in real applications

14.1 Digital photography

A camera sensor samples light from a real scene. The image pipeline may include:

- sensor sampling,
- demosaicing,
- white balance,
- noise reduction,
- color correction,
- compression,
- storage as JPEG, HEIF, RAW, or another format.

Computer graphics and digital photography meet in areas such as HDR imaging, computational photography, and augmented reality.

14.2 Games and animation

Game graphics involve many image representations:

- textures,
- normal maps,
- frame buffers,
- shadow maps,
- depth buffers,
- sprite sheets,
- environment maps.

Even a 3D game is full of 2D images used as data.

14.3 Medical imaging

Medical images often require higher precision than ordinary photographs. CT, MRI, ultrasound, and microscopy images may use specialized formats and bit depths.

In medical visualization, a pixel value may represent tissue density, signal strength, or other measured quantities rather than ordinary color.

14.4 Remote sensing and GIS

Satellite images often contain multiple spectral bands beyond visible RGB. A pixel may store infrared, thermal, or other sensor data.

False-color visualization helps humans interpret these invisible measurements.

14.5 User interfaces

Buttons, icons, fonts, shadows, transparency, and animations all depend on image representation. A modern app screen is built by compositing many visual layers into final pixels.

15. Common beginner mistakes

Mistake 1: Confusing image size and file size

Image size usually means pixel dimensions, such as 1920×1080 .

File size means storage size, such as 500 KB or 5 MB.

A large image can have a small file size if it is highly compressed. A small image can have a large file size if it is uncompressed or has high bit depth.

Mistake 2: Thinking more resolution always means better quality

Resolution matters, but it is not the only factor. Quality also depends on:

- lens or rendering quality,
- compression,
- bit depth,
- color accuracy,
- noise,
- display quality,
- viewing distance.

Mistake 3: Treating JPEG as ideal for everything

JPEG is excellent for many photographs, but it is often bad for text, diagrams, line art, screenshots, and images that need transparency.

Mistake 4: Ignoring color spaces

Two images can have the same RGB numbers but look different if interpreted in different color spaces or displayed on differently calibrated devices.

For this introductory course, we will not go deeply into color management, but we should know that RGB values do not exist in complete isolation.

16. Worked examples

Example 1: Memory requirement of an RGB image

Find the memory required for an uncompressed 800×600 RGB image using 8 bits per channel.

```
Width = 800
Height = 600
Pixels =  $800 \times 600 = 480,000$ 
Bits per pixel = 24
Bytes per pixel =  $24 / 8 = 3$ 
Memory =  $480,000 \times 3 = 1,440,000$  bytes
```

So the image requires about **1.44 MB** in decimal units, or about **1.37 MiB** in binary units.

Example 2: Number of colors in 16-bit RGB 5-6-5

Some systems use 16 bits per pixel, often divided as:

```
Red    = 5 bits
Green  = 6 bits
Blue   = 5 bits
```

Number of colors:

```
 $2^5 \times 2^6 \times 2^5 = 32 \times 64 \times 32 = 65,536$  colors
```

Why does green get 6 bits? Because human vision is relatively sensitive to green detail.

Example 3: Indexed image size

Suppose a 1024×1024 image uses an 8-bit palette index per pixel and a palette of 256 RGB colors.

Pixel index storage:

$$1024 \times 1024 \times 1 \text{ byte} = 1,048,576 \text{ bytes}$$

Palette storage:

$$256 \text{ colors} \times 3 \text{ bytes} = 768 \text{ bytes}$$

Total:

$$1,048,576 + 768 = 1,049,344 \text{ bytes}$$

This is much smaller than direct 24-bit RGB storage:

$$1024 \times 1024 \times 3 = 3,145,728 \text{ bytes}$$

17. Mini activities

Activity 1: Pixel inspection

Open any image in an image editor or programming environment and inspect a pixel value. Ask:

- What is the coordinate of the pixel?
- What are its RGB values?
- Is there an alpha value?
- Does the value match what you visually expected?

Activity 2: Format comparison

Save the same image as JPEG and PNG. Compare:

- file size,
- sharpness of edges,
- compression artifacts,
- transparency support,
- suitability for photographs vs diagrams.

Activity 3: Palette thinking

Imagine a game screen that uses only 16 colors. Design a palette for:

- sky,
- grass,
- water,
- road,
- characters,
- shadows,
- highlights.

This activity shows that representation constraints can become creative design constraints.

18. Summary

Lecture 2 summary: image representation



The key points of this lecture are:

1. A raster image is a rectangular grid of pixels.
2. Each pixel stores numerical attributes such as color, intensity, opacity, or depth.
3. Sampling determines spatial resolution.
4. Quantization determines the number of representable value levels.
5. RGB is the common additive color model for displays.
6. Indexed color uses a lookup table to map pixel indices to actual colors.

7. Bit depth affects quality, memory, and smoothness.
 8. Monitors and printers reproduce images using different physical processes.
 9. Image file formats are designed for different trade-offs.
 10. Image representation is the foundation for later topics such as scan conversion, anti-aliasing, rendering, texture mapping, and animation.
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Short questions

1. What is a pixel?
 2. What is the difference between sampling and quantization?
 3. Why does a 1920×1080 image contain about 2 million pixels?
 4. How many colors can be represented by 24-bit RGB?
 5. What is the purpose of a lookup table in image representation?
 6. Why is JPEG usually better for photographs than diagrams?
 7. Why is PNG useful for icons and screenshots?
 8. What is the difference between RGB and CMYK?
 9. What is an alpha channel?
 10. Why can the same image look different on two different displays?
-

Exam-style questions

Question 1

An image has resolution 1280×720 . Each pixel is stored using 24-bit RGB. Calculate the uncompressed memory requirement in bytes and MiB.

Question 2

Explain direct color coding and indexed color coding. Mention one advantage and one disadvantage of indexed color.

Question 3

Distinguish between sampling and quantization using image representation examples.

Question 4

Why is RGB suitable for display monitors but CMYK more suitable for printers?

Question 5

A grayscale image uses 6 bits per pixel. How many intensity levels can it represent? What visual problem may occur if the number of intensity levels is too small?

Further reading keywords

Search for these terms:

- raster image,
- pixel grid,
- RGB color model,
- indexed color,
- color lookup table,
- bit depth,
- gamma correction,
- image compression,
- JPEG artifacts,
- PNG transparency,
- Mandelbrot set,
- display subpixels,
- CMYK printing.